

On the same wavelength

Words, brainwaves and HPC

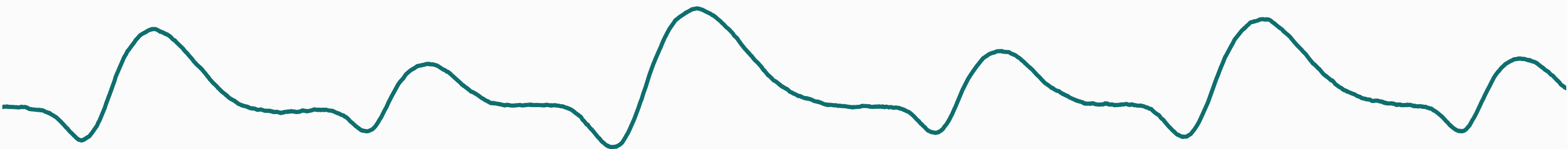
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Oxford Research Computing Community Event

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One power analysis that never finished

ACT 1 — WORDS

Six months

of high-performance computing (HPC) on one question — how many participants does this study need? The largest samples never finished.

- **Cost grows faster than the sample.** 200 simulated data sets take **18 minutes** at 100 participants and **43 hours** at 2,000 — twenty times the participants, over a hundred times the bill.
- **The top of the grid never came back**, and the paper said so at the time. The priming curves stop at 800 participants, so its larger-sample claims are **extrapolation**.

Compute was not a delay here; it was a cap. What finished set what the paper could claim.

On Lancaster’s facility, during my PhD. Not ARC. More hardware would not have rescued it: two of those six months were running, four were waiting.

Thesis reference in the backup slides. Batch timings read back out of the saved `simr` objects.

What the analysis is, and why it has no closed form

ACT 1 — WORDS

- **The question.** Does a word's meaning come from the company it keeps in text, or from how it looks and feels?
- **The data.** One observation is one person, one word, one reaction time — **345,666** in Study 1.
- **The model.** **20 fixed-effect terms and 9 random-effect terms**, per participant *and* per word, because people differ and words differ.
- **The simulation loop.** Simulate new responses from the fitted model, refit the whole thing, test the coefficient.
- **402 R scripts**, submitted one at a time. The same job today is one line: `--array=0-55`.

No formula gives the power of a model like this. So you refit it: 200 times, per effect, per sample size — about 80,000 refits in all.

Sample sizes and model terms from the thesis methods and its `lmerTest` script. Of the 402 scripts, 249 result objects survive.

12,000 core-hours, or 1,150

ACT 1 — WORDS

- **376,054 reaction times.** Redoing that design at full scale: about **12,000 core-hours** — 500 days on one core.
- **One change does most of it:** score all seven effects from **one** set of simulations. Sevenfold, and exact — the same fits, read seven ways.
- **Two more exact steps reach about 1,150** — power is a success count, so chunks add up; and a plainer fit for `simr`, another 1.4×.

>2,000

participants under the archived curves, never reaching 80% power
— an eight-in-ten chance of detecting a real effect



46

participants at full scale,
for the visual-strength effect

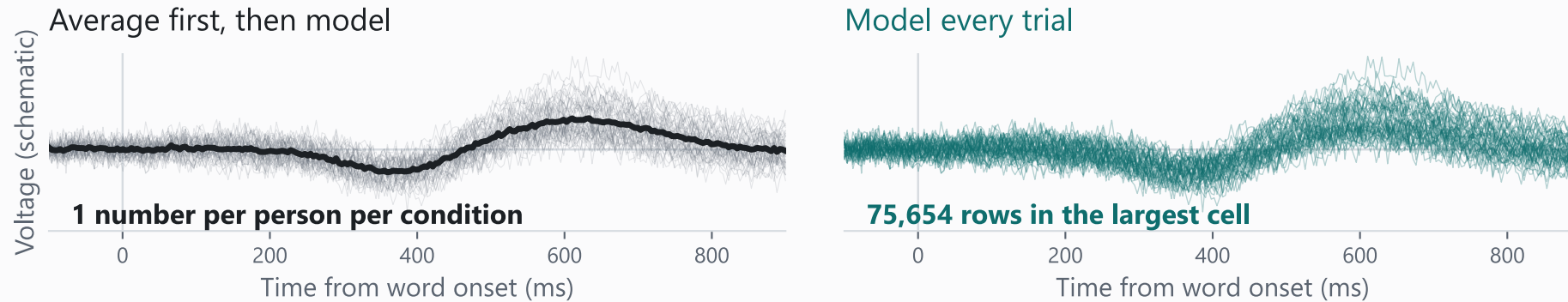
46 is an **analytic screen**, not a `simr` result. It agreed with `simr` on every effect about whether a crossing exists, but read about 40% low where checked. **A bracket, not an answer.**

Algebra, not hardware: 12,000 core-hours became 1,150, and the same algebra moved the sample size.

Measured on the full data set: 199 s per simulate-refit-test at 210 participants, 577 s at 420. The reanalysis is in preparation.

Why EEG became cluster work: we stopped averaging

ACT 2 — BRAINWAVES



- **The recording.** Electroencephalography (EEG): a voltage at 32 electrodes every 2 ms while someone reads.
- **The conventional move.** Average the surviving trials per condition per person — about 31 of the 48 presented — into one clean waveform.
- **What we do instead.** Model every trial: 728,352 rows.
- **Correlated by-participant and by-item slopes:** one person's effects may move together, and one word's. The legacy forbade it, to make the old software converge.
- **32,000 posterior draws per fit** — 32,000 plausible answers, not one estimate.
- **102 fits, all converged, zero divergent transitions.** Worst R-hat 1.0072, against a gate set before fitting: R-hat below 1.01.

One choice drives every core-hour here: 728,352 single-trial rows, not 23,000 averages.

Numbers from committed result files. Study design: González Alonso et al. (2025), preregistration osf.io/tjr54; the single-trial Bayesian analyses shown here are in preparation. 96 of the 102 are event-related-potential (ERP) fits. The figure is schematic, not data: simulated trials with an N400-P600 shape.

The array index is the experimental design

ACT 2 — BRAINWAVES

```
#SBATCH --job-name=p1_fit_erp
#SBATCH --partition=long
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem-per-cpu=12G
#SBATCH --time=5-00:00:00
#SBATCH --array=0-17

source _shared/hpc/arc_env.sh
export LES_THREADS_PER_CHAIN=2
```

- **8 CPUs is 4 chains × 2 threads each** — four chains, because convergence is judged by comparing them.

	200–500 ms		300–600 ms		400–900 ms	
Gender agreement	0	1	2	3	4	5
Differential object marking	6	7	8	9	10	11
Verb–object number agreement	12	13	14	15	16	17

teal = lateral · pale = midline · enumerated property-major, then window, then macroregion

- **18 tasks is 3 grammatical properties × 3 time windows × 2 scalp macroregions.** `--mem-per-cpu` multiplies: 96 GB per task.
- Upstream, extraction asks for **4 × 48 GB = 192 GB**, “because the legacy merge loads the full single-trial EEG for one property at a time.”

192 GB to prepare the data, 96 GB to model it: memory, not cores, sizes these jobs.

03_fit_erp.slurm, SBATCH block with the log paths elided, and its 18-row PARAMS block; 01_extract_erp.slurm.

Two thirds of the array died in the compiler

ACT 2 — BRAINWAVES

That array never reached the sampler. Two thirds of its tasks died while building the model:

```
g++: internal compiler error: Segmentation fault signal terminated program cc1plus
```

An internal compiler error: GCC 14.2.0, the compiler inside `R/4.5.1-gfbbf-2025a`, giving up on Stan’s `reduce_sum` templates. Measured on a compute node:

Build	Result
threading + <code>-O3</code>	ICE after 19 s
threading + <code>-O2</code>	ICE after 4 s
threading + <code>-O1</code>	builds in 90 s — chosen
no threading + <code>-O3</code>	builds in 82 s

Two fixes cleared the crash; the faster one would have changed the posterior.

99_diagnose_ice.slurm, 99b_test_02.slurm, _shared/hpc/arc_env.sh.

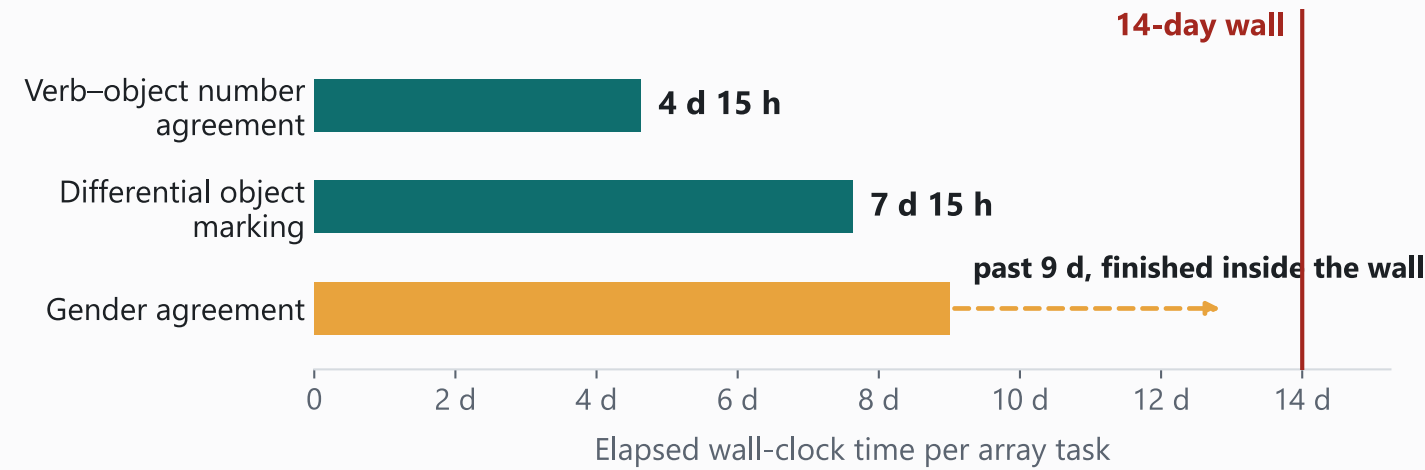
- **Peak memory under 2 GB** against a 96 GB allocation. **A compiler bug, not a resource limit.**
- **The fix is one line**, appended to `$CMDSTAN/make/local` by every job:

```
CXXFLAGS_OPTIM = -O1
```
- **Dropping threading would have worked too.** But threads add up the likelihood in pieces, and a different order of adding changes **the exact posterior draws**. The optimisation level does not.
- **Refitted with that line, and no recurrence since.**

The machine learning is CPU-bound, and it is the longest job

ACT 2 — BRAINWAVES

- **The classifier.** Decode *Grammatical* against *Ungrammatical* from all the electrodes at once, at each time point, with a ridge-regularised linear discriminant, scored by area under the curve (**AUC**), where 0.5 is chance.
- **The leakage guard.** Leave one whole participant out of every fold, so no one person sits on both sides of a split. Chance is measured, not assumed: shuffle the labels within participant and re-run the entire decoder, **1,000 times**.
- **What that costs.** For one property, 300 time points × 63 participants × 1,001 fits ≈ **18.9 million** discriminants, each on a 31 × 31 covariance.



`--array=0-2` — one task per property, 8 CPUs × 32 GB = **256 GB**, and the memory goes on rebuilding the trial × channel × time tensor, not on the decoding.

What nine days bought: grammaticality decodable from about **380 ms** after the critical word, at an AUC of 0.53, above the permutation null.

The machine learning never touches a GPU: 18.9 million tiny fits, none big enough to fill one.

09_run_decoding.R, 08_decoding.slurm. 18.9 M is my own arithmetic from sourced parts. Window and AUC from the committed gender-agreement time course. Verb-object number agreement yields nothing significant. Grepping the EEG repositories for `gres=gpu`, `cuda` and `torch` returns only lockfile metadata.

Some data cannot go to the cloud

ACT 3 — SPEECH

“Audio recordings and transcriptions ... **never leave the institution** under which the recording was made, are **never transmitted to a remote service**, and are **never used to train any model**.”

The workflow’s own privacy guarantee. The recordings are children’s focus groups and research interviews: special-category data under Article 9 of the UK General Data Protection Regulation (GDPR), the law’s strictest tier.

Beyond the law, **opacity about model versions** makes a cloud transcription unreproducible. Nor is there much accuracy to trade — Landesvatter et al. (2025) found Google Cloud beaten by almost every open-weight system tested.

Where the documentation stops: the ARC user guide covers connecting, storage, quotas and backups. Special-category data is not in it — what may be held, where, or who to ask first.

The recordings cannot leave the institution, so the compute has to come to them.

Privacy guarantee: PYANNOTE_SETUP_GUIDE.md L11. Documentation gap: the ARC user guide’s own contents, arc-user-guide.readthedocs.io, read September 2026. Landesvatter et al. (2025) in the backup references.

Whisper and pyannote as an ARC job array

ACT 3 — SPEECH

```
#SBATCH --array=0-4
#SBATCH --partition=short
#SBATCH --clusters=htc
#SBATCH --cpus-per-task=4
#SBATCH --mem=32G
#SBATCH --gres=gpu:1
```

- `openai/whisper-large-v3` on GPU, with `pyannote/speaker-diarization-3.1` labelling who spoke when on the same device. One array task per corpus, resubmitted per model: **3 models × 5 corpora × 16 text-cleanup conditions = 240 cells.**

Large-v3: **6.2% word error rate on read speech, 32% on meeting audio** — roughly one word in three to correct. Same model; the audio decides.

```
JID=$(sbatch --parsable hpc/submit_sweep.sh)
sbatch --dependency=afterok:${JID} hpc/aggregate_sweep.sh
```

Aggregation runs only if decoding succeeded: no GPU, 8 GB, half an hour.

The GPU decodes each corpus once; a seventeenth condition costs no GPU time. Don't spend a GPU allocation on string comparison.

`submit_sweep.sh` and `aggregate_sweep.sh`, log paths elided. Word error rate from the committed `sweep_aggregate.csv`, macro-averaged over LibriSpeech test-clean and the AMI Meeting Corpus individual-headset-microphone split (Carletta et al., 2006), CC BY 4.0.

Two checks I could afford to run

ACT 3 — SPEECH

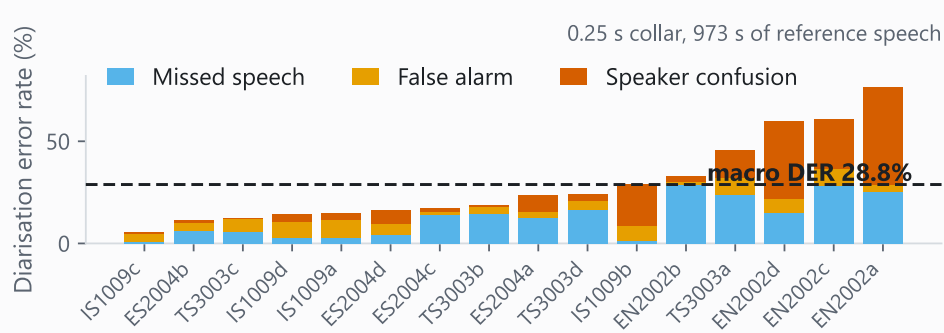
1 — A fix that halved the end-boundary error

Over 499 AMI meeting utterances, Whisper’s starts are near frame-accurate; its **ends are not**. Silero, a 1 MB voice-activity detector, re-cuts each end at the last voiced frame.

Boundary error	before	after
End, mean absolute	127.1 ms	63.9 ms
End, root-mean-square	460.5 ms	253.6 ms
Start, mean absolute	8.7 ms	8.7 ms
Start, root-mean-square	70.9 ms	70.9 ms

Root-mean-square error punishes outliers: 460 ms is outside the 0.25 s collar against, 254 ms inside. The start rows are **bit-identical**.

2 — A change that made diarisation worse



Lower pyannote’s clustering threshold to 0.60 and diarisation gets **worse**: 28.8% → 31.9% of speech time mislabelled. Clustering only renames speakers, so missed speech and false alarm stay **bit-identical**.

16 mini-meetings, not the full test set, and that average hides a spread from 5.6% to 76.7%.

Cheap re-runs change what you are willing to verify — and to publish.

timestamp_summary.json versus evaluation_vad_refine/ (one utterance of 500 skipped); der_summary.json versus evaluation_clustering_0.60/. Both runs: one short GPU job on an H100 node.

Where the GPU is, and where it is not

	What	Where
GPU, inference	Whisper large-v3, distil-large-v3 and medium; pyannote 3.1 diarisation; Silero VAD; six Pythia language models, 70m–2.8b parameters	ARC <code>htc</code>
GPU, training	Fine-tuning <code>pyannote/segmentation-3.0</code> . The only model I train , because far-field diarisation runs near 57% error — the sound of a real focus group.	ARC <code>htc</code>
CPU, machine learning	Leave-one-participant-out ridge linear-discriminant decoding; projection-predictive variable selection; brms and Stan throughout	ARC
No GPU, on purpose	<i>“Do not add <code>--gres=gpu</code> to any SLURM script. Stan/CmdStan does not benefit from GPU for the ordinal mixed models used here.”</i> Continuous integration fails the build if a GPU directive appears.	ARC

An unconstrained GPU job died in **14 seconds** on a Skylake node — “*your machine doesn’t support AVX512_CLX*”. The fix was a **CPU**-generation constraint on a **GPU** job.

The GPUs here run other people’s models as measuring instruments, not as things to train.

`submit_benchmark_wer_gpu.sh`, `submit_finetune_pyannote.sh`, `09_run_decoding.R`, `08_projpred_selection.R`, the CLAPS ARC workflow doc, and `fit-surprisal-extended.slurm`.

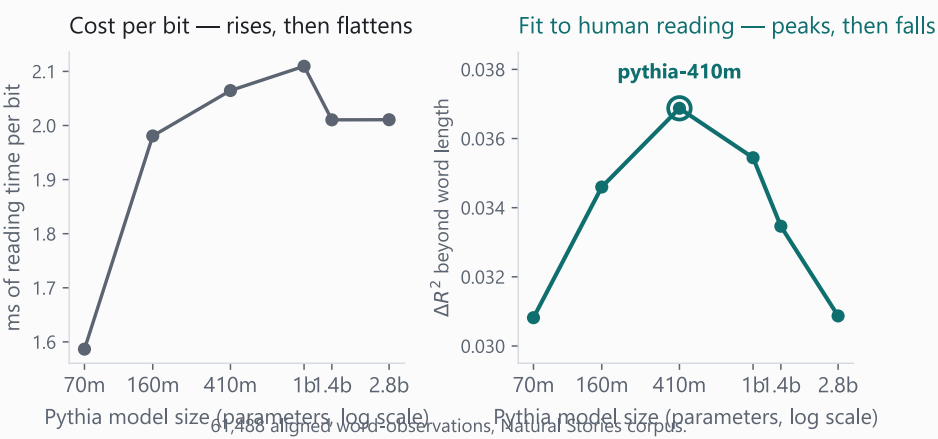
What the cluster made possible

Words. Six Pythia models scored in **one GPU job**: how unexpected each word was, against how long people took to read it. **Inverse scaling**: past a certain size, a better language model predicts *people* worse.

Brainwaves. Chance measured rather than assumed: 1,000 re-runs of the decoder for each property, including the one that came back null.

Speech. A word error rate that depends on what you agree to throw away. The same transcripts score 6.2% on read speech and 2.3% under the canonical normaliser, which discards the punctuation, casing and fillers my questions need.

Design. A sample-size requirement bracketed near 46 where the old one said more than 2,000 — **before** anyone is recruited.



In all four, the cluster changed which question I could ask. Not faster — different.

`fit-surprisal-extended.slurm` and the chapter-19 freeze output. Paired bootstrap over words: pythia-410m beats pythia-2.8b in 99% of 500 resamples, 95% confidence interval on ΔR^2 [0.0010, 0.0121]. Full references in the backup slides.

What would help most

What I'd ask for

- **A documented route for special-category data.** The user guide covers storage and backups, not what may be held or who to agree it with. That conversation decides what social-science work I can bring here.
- **The array-size cap, written down.** `htc` capped one of my arrays at 1,000 tasks: 4,800 submitted, 1,000 ran, and a preregistration carries 37 to 48 converged simulations a cell instead of 300. The scheduling page documents partitions and walltimes, but not this.

What I can offer

- **The compiler bug, upstream.** A five-line reproducer and a one-line fix for the internal compiler error in `R/4.5.1-gfbf-2025a`, sent this week.
- **A case study.** My field's standard toolkits — EEGLAB, FieldTrip, MNE, SPM, Praat — are none of them in the module list. I would write up what a social-science user installs.

Nothing here is a request for more compute. It is one conversation, one limit written down, and a bug report.

Acknowledgement, as ARC asks: "The authors would like to acknowledge the use of the University of Oxford Advanced Research Computing (ARC) facility in carrying out this work. <https://doi.org/10.5281/zenodo.22558>"
pablo.bernabeu@education.ox.ac.uk · ORCID 0000-0003-1083-2460 · github.com/pablobernabeu · Transcription workflow: doi.org/10.5281/zenodo.17624830

Backup slides

Material for questions.

Two operational lessons, both mine

- A node was drained for a **datacentre thermal event** and Slurm requeued the task correctly. Eleven of thirteen blocks were complete, and nothing was on disk. Another run was lost to a **NODE_FAIL**. Between them, more than a fortnight, all of it recoverable at the cost of one design decision.
- **The fix:** checkpoint every 50 permutations, caching **.Random.seed** alongside, so a resumed run is bit-identical to an uninterrupted one. In the code's own words, *“an artefact must not depend on how often the scheduler killed the job.”*
- **Never **scp** a script that a running job is executing.** **Rscript** parses one expression at a time, holding a byte offset into the file, so overwriting shifts the content underneath it. An array element five days into its run wrote every artefact, then died on a parse error, breaking the **afterok** dependency downstream.
- **Hence a watcher job** — 20 minutes, 4 GB — submitted with **--dependency=afterany**, precisely because that array is expected to exit non-zero on complete success.
- **A chaining lesson.** A three-minute cross-property analysis, chained **afterok** on the whole decoding array, sat queued behind a fortnight of permutation runs.

Anticipated questions

How much compute have you used? It varies enormously by project, and the total is never the interesting number. What binds is memory, a toolchain or a wall-clock limit, essentially never raw throughput.

Why not run Stan on a GPU? For these models it does not help. The bottleneck is posterior geometry and the *number* of fits, not dense linear algebra. The project's own rule says so, and CI enforces it.

Would a GPU help the LDA decoding? Possibly, but the cost is 18.9 million *small* fits and the permutation null is embarrassingly parallel across time points and folds. An array of CPU tasks fits the shape better. I have not benchmarked the alternative, so I will not claim it.

Isn't 1,000 permutations overkill? No. At 200 the p -floor is $1/(B+1) = .00498$, which exceeds α/m for the confirmatory cluster family. The code refuses its own earlier artefacts on exactly that ground.

Do you let an agent submit jobs? It polls `squeue` and `sacct` read-only and proposes the next stage; submission is mine. Standard QoS only, never the priority flag, admin holds never released, nothing cancelled without confirming supersession.

What is the word error rate on your own interviews? Unknown, and I will not guess. A ground truth means a second human transcript, and two transcribers disagree with each other. The benchmark corpora tell me how much correcting to expect; they do not tell me how accurate my transcripts are. That is why the workflow outputs a draft a researcher corrects.

Did the pyannotate fine-tuning improve anything? Not yet. The job is written and resourced against a far-field baseline near 57% diarisation error, and it has not been run to a result I can report.

Why `-O1` rather than dropping threading? Because dropping threading changes `reduce_sum`'s reduction order and therefore the exact posterior draws. The optimisation level does not.

Have you reported the compiler bug upstream? The reproducer is five lines and it goes this week — a pointer to the right route would help.

Is this not BMRC's problem? BMRC is a separate service. Education is in Social Sciences, so ARC is the service I am entitled to and the one I use.

Could the 14-day wall be shorter? Fourteen days is what I asked for, not the partition's ceiling. For the permutation null I would not go lower, but a fortnight backfills badly, and I would rather checkpoint and request less than hold a reservation that size.

References

Bernabeu, P. (2025). *Secure and scalable speech transcription for local and HPC* (Version 1.0.0) [Computer software]. Zenodo. <https://doi.org/10.5281/zenodo.17624830>

Bernabeu, P. (2022). *Language and sensorimotor simulation in conceptual processing: Multilevel analysis and statistical power* [Doctoral thesis, Lancaster University]. Lancaster University research directory. <https://doi.org/10.17635/lancaster/thesis/1795>

Biderman, S., Schoelkopf, H., Anthony, Q., Bradley, H., O'Brien, K., Hallahan, E., Khan, M. A., Purohit, S., Prashanth, U. S., Raff, E., Skowron, A., Sutawika, L., & van der Wal, O. (2023). Pythia: A suite for analyzing large language models across training and scaling. *Proceedings of Machine Learning Research*, 202, 2397–2430. <https://proceedings.mlr.press/v202/biderman23a.html>

Carletta, J., Ashby, S., Bourban, S., Flynn, M., Guillemot, M., Hain, T., Kadlec, J., Karaiskos, V., Kraaij, W., Kronenthal, M., Lathoud, G., Lincoln, M., Lisowska, A., McCowan, I., Post, W., Reidsma, D., & Wellner, P. (2006). The AMI Meeting Corpus: A pre-announcement. In S. Renals & S. Bengio (Eds.), *Machine learning for multimodal interaction* (pp. 28–39). Springer. https://doi.org/10.1007/11677482_3

Futrell, R., Gibson, E., Tily, H. J., Blank, I., Vishnevetsky, A., Piantadosi, S. T., & Fedorenko, E. (2021). The Natural Stories corpus: A reading-time corpus of English texts containing rare syntactic constructions. *Language Resources and Evaluation*, 55(1), 63–77. <https://doi.org/10.1007/s10579-020-09503-7>

González Alonso, J., Bernabeu, P., Silva, G., DeLuca, V., Poch, C., Ivanova, I., & Rothman, J. (2025). Starting from the very beginning: Unraveling third language (L3) development with longitudinal data from artificial language learning and EEG. *International Journal of Multilingualism*, 22(1), 119–142. <https://doi.org/10.1080/14790718.2024.2415993>

Landesvatter, C., Behnert, J., & Bauer, P. C. (2025). Comparing speech-to-text algorithms for transcribing voice data from surveys. *Public Opinion Quarterly*, 89(4), 1154–1166. <https://doi.org/10.1093/poq/nfaf056>

Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2023). Robust speech recognition via large-scale weak supervision. *Proceedings of Machine Learning Research*, 202, 28492–28518. <https://proceedings.mlr.press/v202/radford23a.html>

Richards, A. (2015). *University of Oxford Advanced Research Computing* [Technical note]. Zenodo. <https://doi.org/10.5281/zenodo.22558>

Software and model versions are given on the slides that use them. Full list, with links, in the repository.